

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

# Read image
image = cv2.imread("image.jpg", cv2.IMREAD_GRAYSCALE)

# Resize image
image = cv2.resize(image, (100, 100))

# Convert image into 1D feature vector
# Original: data = image.reshape(1, -1)
# Fix: Treat each row of the image as a sample
data = image

print("Original Feature Dimension:", data.shape[1])

# Different PCA component values
components = [10, 25, 50, 75, 100]

plt.figure(figsize=(15, 8))

for i, n in enumerate(components):

    # Apply PCA
    pca = PCA(n_components=n)

    transformed = pca.fit_transform(data)

    # Reconstruct image
    reconstructed = pca.inverse_transform(transformed)

    # The reconstruction will have the original image shape
    # reconstructed = reconstructed.reshape(100, 100)

    # Display result
```

```
plt.subplot(2, 3, i + 1)
plt.imshow(reconstructed, cmap="gray")
plt.title("Components = " + str(n))
plt.axis("off")

print("\nComponents:", n)
print("Reduced Dimension:", transformed.shape)
print("Explained Variance:",
      np.sum(pca.explained_variance_ratio_))

plt.tight_layout()
plt.show()
```

Original Feature Dimension: 100

Components: 10

Reduced Dimension: (100, 10)

Explained Variance: 0.8828829246593467

Components: 25

Reduced Dimension: (100, 25)

Explained Variance: 0.9661724215940527

Components: 50

Reduced Dimension: (100, 50)

Explained Variance: 0.9970268296404593

Components: 75

Reduced Dimension: (100, 75)

Explained Variance: 0.9999404567172828

Components: 100

Reduced Dimension: (100, 100)

Explained Variance: 1.0

Components = 10



Components = 25



Components = 50

